

Normalization

// Regular entities

Person (PersonID, SSN, FirstName, LastName, Sex, BirthDate, DeathDate, Picture)

Anomaly:

// DeathDate NULL value?

Person (PersonID, SSN, FirstName, LastName, Sex, BirthDate, Picture)

DeadPerson (PersonID, DeathDate)

1NF:

Composite:	Not found
Multivalued:	Fixed
Nested relations:	Not found

2NF:

Total:	Not found
Partial:	Not found

3NF:

Transitive:	Not found
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BCNF:

Transitive:	Not found
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Home (AddressID, Province, City, Street, BuildingNumber)

1NF:

Composite:	Not found
Multivalued:	Not found
Nested relations:	Not found

2NF:

Total:	Not found
Partial:	Not found

3NF:

Transitive:	Not found
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BCNF:

Transitive:	Not found
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Organization (OrganizationID, OrganizationName)

1NF:

Composite:	Not found
Multivalued:	Not found
Nested relations:	Not found

2NF:

Total:	Not found
Partial:	Not found

3NF:

Transitive:	Not found
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BCNF:

Transitive:	Not found
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// Weak entities

Person_Achievement (PersonID, AchievementID, Type, Date) // 1:N relationship

1NF:

Composite:	Not found
Multivalued:	Not found
Nested relations:	Not found

2NF:

Total:	PersonID, AchievementID → Date
Partial:	AchievementID → Type // Type is a candidate key

Person_Achievement (PersonID, AchievementID, Date)

Achievement (ArchievementID, Type)

3NF:

Transitive: Not found

BCNF:

Transitive: Not found

Final:

Person_Achievement (PersonID, ArchievementID, Date)

Achievement (ArchievementID, Type)

// M:N relationship

Person_LivesIn_Home (PersonID, AddressID, StartDate, EndDate)

Anomaly:

// Similar issues with Person-DeathDate

1NF:

Composite: Not found

Multivalued: Not found

Nested relations: Not found

2NF:

Total: Fixed

Partial: Not found

3NF:

Transitive: Not found

BCNF:

Transitive: Not found

Person_EducatedIn_School (PersonID, OrganizationID, StartDate, EndDate, Degree)

Anomaly:

// Similar issues with Person-DeathDate

1NF:

Composite: Not found

Multivalued: Not found

Nested relations: Not found

2NF:

Total: Fixed

Partial: Not found

3NF:

Transitive: Not found

BCNF:

Transitive: Not found

Person_WorkedIn_Company (PersonID, OrganizationID, StartDate, EndDate, JobPosition, Income)

Anomaly:

// Need to set JobPosition as an entity? Easy to modify the Job description.

// Maybe should delete Income, because Income may change with time during a single JobPositon.

// Or we just store approximate records

1NF:

Composite: Not found

Multivalued: Not found

Nested relations: Not found

2NF:

Total: Fixed

Partial: Not found

3NF:

Transitive: Not found

BCNF:

Transitive: Not found

// Recursive relationship

Father_FatherOf_Child (Father_PersonID, Child_PersonID, IsGenetically)

Monther_MontherOf_Child (Monther_PersonID, Child_PersonID, IsGenetically)

Husband_SpouseOf_Wife (Husband_PersonID, Wife_PersonID, StartDate, EndDate)

Anomaly:

// Similar issues with Person-DeathDate

// Multivalued attributes

Person_MiddleName (PersonID, MiddleName)

// Fixed in 1NF

Person_UsedName (PersonID, UsedName)

// Fixed in 1NF

Final:

Person (PersonID, SSN, FirstName, LastName, Sex, BirthDate, DeathDate, Picture)

Home (AddressID, Province, City, Street, BuildingNumber)

Organization (OrganizationID, OrganizationName)

Achievement (ArchievementID, AchievementDescription)

AchievementHistory (PersonID, ArchievementID, AchievementDate)

HomeHistory (PersonID, AddressID, StartDate, EndDate)

EducationHistory (PersonID, OrganizationID, StartDate, EndDate, Degree)

WorkHistory (PersonID, OrganizationID, StartDate, EndDate, JobPosition, Income)

FatherOf (Father_PersonID, Child_PersonID, IsGenetically)

MontherOf (Monther_PersonID, Child_PersonID, IsGenetically)

SpouseHistory (Husband_PersonID, Wife_PersonID, StartDate, EndDate)

Person_MiddleName (PersonID, MiddleName)

Person_UsedName (PersonID, UsedName)